Storage: New challenge for renewables

Renewables accounted for a remarkable 83 per cent of the world's power generation capacity addition in 2022, spearheaded by wind and solar. However, the mainstreaming of renewables has cast the spotlight on their inherent challenges — that they are intermittent and variable. The solution is energy "storage", which is set to grow at 23 per cent per annum. While China and the US are expected to drive around 60 per cent of these additions, India presents an intriguing scenario. Despite being projected to have the world's third-largest renewable energy capacity by 2030, India is likely to be fifth in terms of energy-storage installation.

In May 2021, the Indian government announced a ₹18,000 crore PLI (production-linked incentive) scheme for advanced cell chemistry battery manufacturing, hoping to draw in foreign and domestic investment of ₹45,000 crore.

INFRATALK

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In August 2023, the Ministry of Power issued the National Framework for Promoting "Energy Storage Systems" (ESS). This framework addresses in detail procedures and incentives across five subsectors of ESS. They are (i) storage with generation of renewable energy; (ii) storage with transmission; (iii) storage with distribution; (iv) standalone ESS operating independently as "merchant units"; and (v) storage for ancillary and balancing services.

Following this framework, in September 2023, India approved a ₹3,760 crore viability gap fund aimed to encourage the production of "Battery Energy Storage Systems" (BESS) for renewable energy storage. Under the scheme, the government committed itself to providing financial support of up to 40 per cent of the capital cost of BESS projects totalling 4,000 MW hours (MWh) till 2030-31. The government reiterated that this was in keeping with its desire to reduce the levelised cost of storage to ₹5.50-6.60 per KWh.

Further, the Ministry of Power has indicated the directive could be expected anytime now to have renewable projects with more than 5 MW install energy storage system (ESS) for minimum 5 per cent of their capacity.

In mid-December 2023, India's Ministry of Environment categorically stated at COP28, Dubai, that until storage (and related "abatement" technologies) became viable, the country could not commit itself to definitive time-lines to phase out fossil fuels, particularly coal.

Demonstrating its commitment to the global climate action agenda, India has set a target to achieve a 500 GW renewable energy generation capacity by 2030, primarily supported by 292 GW of solar power. With anticipated misalignment between solar peak hours and demand peak hours, ESS will be critical.

The most common types of storage are BESS and PSP (pumped storage projects). Green hydrogen and ammonia are also considered alternative forms of storage, currently at various stages of development. Coupling ESS with solar and wind generation is paramount not just for ensuring energy supply reliability but also for providing a variety of required interventions for frequency response and voltage support towards grid quality improvement.

The Central Electricity Authority has projected a requirement of 60 GW of energy storage capacity by 2030; ie 12 per cent of the projected 500 GW of renewable capacity planned to be set up in the same period. This is to consist of BESS of 42 GW and PSP of 18 GW. Against this projection of 60 GW, the current visibility is about 30 GW.

Lithium is the key mineral ingredient for BESS. At a global level, Chile, Australia, China, Argentina and Brazil are reckoned to be the largest producers of lithium. India's lithium reserves were recently discovered in Jammu & Kashmir, and Rajasthan. Together, these two discoveries are estimated to take care of 90 per cent of India's demand. In spite of this, there is serious concern about lithium availability for batteries, volatility in other material components, high capex requirements for manufacture et al. In spite of various impediments, the power minister, R K Singh, informed Parliament recently that with the supporting environment in place, the cost of BESS was anticipated to be in the range of ₹2.20-2.40 crore per MWh in 2023-26.

The October 28 issue of *The Economist* has a detailed article on how cheap and abundant "sodium" could replace, or at least rival, lithium-based batteries with its version of sodium-ion (Na-ion) technology gaining traction alongside lithium-ion (Li-ion) batteries. If this sodium-based technology can be commercialised soon, it would be a big boost for India, with its vast availability of this salt.

As a viable alternative, India's familiarity with PSP makes it a compelling option for electricity storage, given its mature and proven technology. In essence, PSP harnesses surplus renewable electricity to pump water up and then release it down to hydropower plants, generating electricity as and when demand arises. Thus, the policy focus in the short term is, rightfully, to rapidly operationalise PHS, bolstered by the fact that India has already identified 119 GW of PSP potential.

While the discovered "standalone" storage tariff has been ₹9-11/kWh, the integrated tenders have seen tariffs of ₹3-7/kWh. It may be clearly inferred that integrated tenders enjoy the advantage of combining and optimising solar and wind capacity with storage capacity to match demand on a round-the-clock-basis, or in accordance with the dispatch requirements.

With various aspects of the energy storage ecosystem demanding attention, implementing a supportive framework is pivotal. Some of the suggestions are:

(i) Scheme of waiver of inter-state transmission charges to be extended till 2030 for electricity generated by ESS. This is currently applicable only for BESS/PHS projects commissioned up to June 30, 2025.

(ii) Rectifying the double-charging framework which treats ESS as a consumer while charging, and as a generator during discharging, incurring network charges twice. Globally, countries like the UK, Japan, and the US have tried resolving this by recognising storage as a generator. In India, the National Framework document labels energy storage as an intermediary element, advocating exemption of input charging power from various network charges. However, a crucial step forward will need a gazette notification to make it binding for states.

(iii) Exemption from open access charges and transmission charges needs to be done by states to promote storage installation.

(iv) Concessional goods and services tax of 5 per cent on grid-scale BESS vis-à-vis 18 per cent currently.

The Central Electricity Authority has projected that by 2047, the requirement of energy storage is expected to increase to 320 GW (90 GW PSP and 230 GW BESS) due to the addition of a larger amount of renewable energy vis-à-vis the net zero emissions targets set for 2070. Thus, the implementation of storage for India's green transition clearly emerges as the new challenge for the renewables vision.

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Disclaimer: Vinayak Chatterjee's earlier piece published on this page on November 22, titled Mundra Port@25: Trailblazing entrepreneurship, *did not make the disclosure that Mr Chatterjee is a part-time advisor to the chairman of Adani Group, Mr Gautam Adani, on strategic issues. The omission of that disclosure is regretted*